

Python with machine learning.

9/11/25 Machine learning with Python. Means using the Python programming language to build computer programs that can learn from data and make predictions or decisions without being explicitly programmed.

Python is the most popular language for ML because it is

- Easy to learn
- Has powerful ML libraries
- Has strong community support
- Works well with data analysis & visualizations

What is programming

Set of instructions are given to machine to perform specific task is called as programming.

What is programming language

To give instruction to machine we use programming language to perform the given task.

Different types or levels of programming.

① Low level programming : These are closest to the machine (hardware). They are fast but hard for humans to read & write.

a) Machine language (level 0)

- written in binary (0s & 1s) • Very difficult for humans
- Directly understood by the computer.

Ex: 100101001010

b) Assembly language (level 1)

- uses mnemonics (short words) instead of binary
- Easier than machine code

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- Needs than machine code
 - Needs an assembler to convert to machine code
- Ex: - MOV A, 05
Add A, B

② Middle-level programming: These are a mix of low-level + high-level features

Ex: C → • Can interact with hardware (low-level)
• Also support complex programs (high-level)

③ High-level programming language: These are close to human language. Easy to learn, easy to write

Ex: Python, Java, C++, JavaScript

features: • Easy syntax • portable
• Requires compiler or interpreter
• Slower than low level, but easier to develop

PYTHON: High level programming, interpreted by line by line, it shows the errors in first line

object oriented programming, indented programming language, dynamically programming language

- Python is a high level, interpreted P.L known for its simplicity and readability. It allows developers to write clear programs for both small & large-scale project. Python is used for:
 - web development
 - data science & ML (pandas, Tensorflow)
 - Automation
 - Software development & more

Why python popular

- Easy to learn
- Community support

- Cross-platform
- Versatile

Python as an interpreted language unlike other compiled languages like C or Java, Python executes the code line by line, which makes debugging easy. Python doesn't require you to compile your code into machine language, the Python Interpreter takes care of it.

Who is the developer of Python?
Guido van Rossum in the year of 1985
officially 1991 Feb 20

2nd version Python 2000 Oct 16

3rd " " 2008 Dec 3

Features of Python

1. Simple syntax: Easy to read & write, similar to English
2. Interpreted: Python executed by line by line
3. Dynamically typed: No need to declare variable types explicitly
4. Object oriented: Support oop like classes and object
5. Rich standard library: Comes with lots of built-in modules & functions
6. Free / Open Source
7. Extensive standard libraries
8. Cross platform compatibility
9. Strong community support
10. Highly versatile